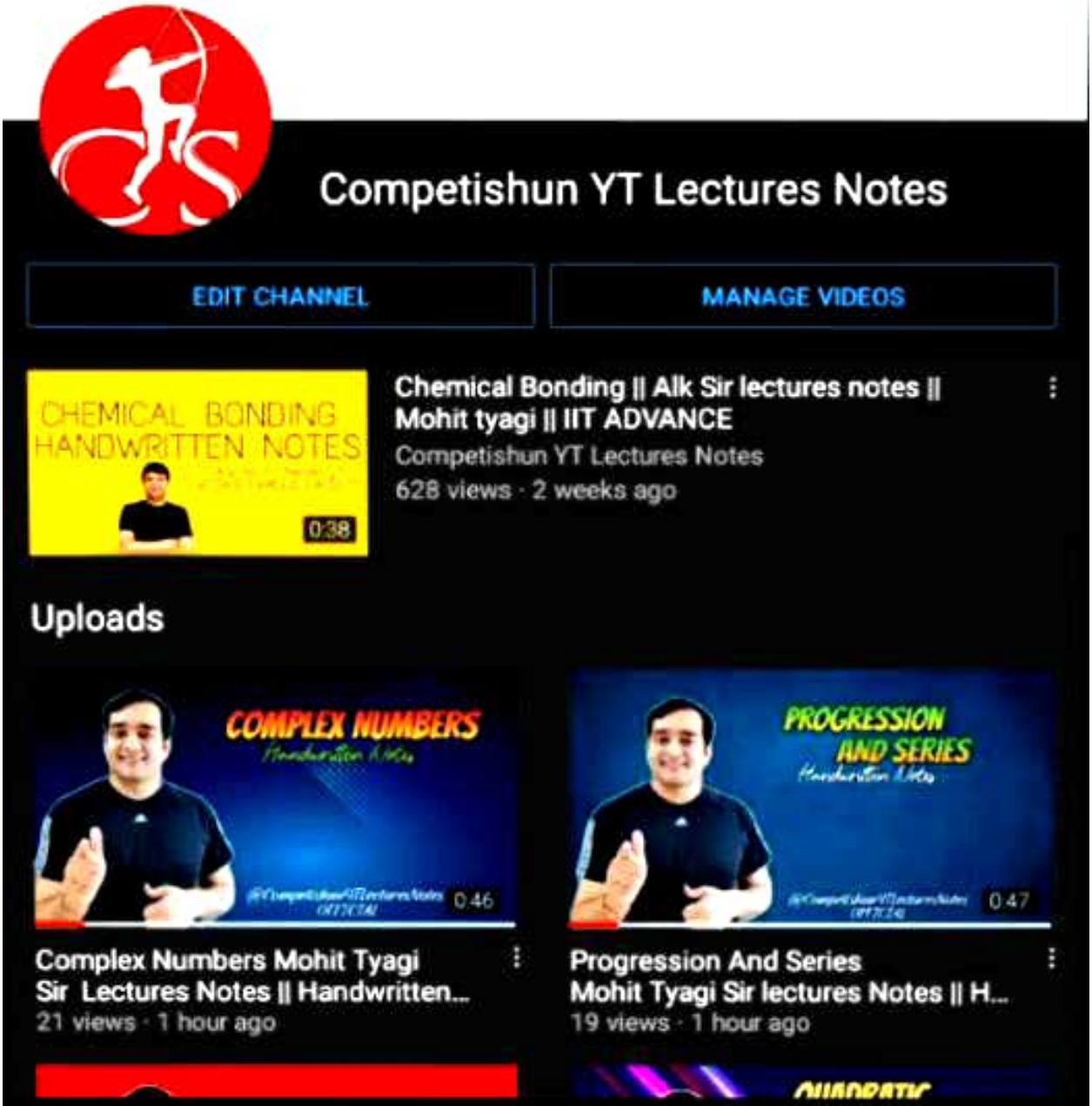


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BIOMOLECULES ⇒ (I) CARBOHYDRATES

↓

Hydrate of carbon

$C_n(H_2O)_n$

- Glucose = $C_6H_{12}O_6 \Rightarrow C_6(H_2O)_6$
- Ribose = $C_5H_{10}O_6 \Rightarrow C_5(H_2O)_5$

} Carbohydrate

- acetic acid $CH_3\text{-}COOH$ $C_2H_4O_2 = C_2(H_2O)_2$
- formaldehyde $C(H_2O)$

} Not carbohydrate yet satisfies the formula

Carbohydrate → { Sucrose $C_{12}H_{22}O_{11}$ } yet does not satisfy the above formula

BIOMOLECULES ⇒ (I) CARBOHYDRATES

- Carbohydrate ⇒ Polyhydroxy aldehyde/Ketone

or

Cyclic hemiacetal/Hemiketal/acetal/Ketal of polyhydroxy carbonyl comp.

or

Compounds which can be hydrolysed to polyhydroxy carbonyl comp.

⇒ Monosaccharides:- Carbohydrate which can not be hydrolysed to small carbohydrate molecules. all satisfy the Cn(H2O)n formula

eg. D. Glucose, D. Fructose, D. Mannose, D. Galactose, D. Ribose etc.

⇒ Disaccharides:- Carbohydrate which can be hydrolysed to two molecules of monosaccharide

eg Maltose, Sucrose, lactose etc.

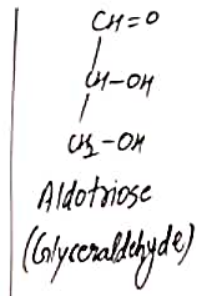
⇒ Polysaccharides:- can be hydrolysed to many molecules of monosaccharides.

eg Starch, Cellulose, Glycogen etc

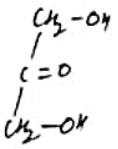
BIOMOLECULES ⇒ (I) CARBOHYDRATES

⇒ General Nomenclature of carbohydrates:-

- Suffix . 'ose' **generally, there are exceptions ex. glycogen**
- on the basis of no. of carbon atoms:- Triose, Tetrose, Pentose, Hexose, Heptose.
- Aldehyde group containing carbohydrate ⇒ Aldose
- Ketonic ⇒ Ketose
no. of carbon	Aldose	Ketose
C_3	Aldotriose	Ketotriose
C_4	Aldotetrose	Ketotetrose
C_5	Aldopentose	Ketopentose
C_6	Aldohexose	ketohexose
C_7	Aldoheptose	ketoheptose



Aldotriose
(Glyceraldehyde)



Ketotriose

except this carbohydrate, all are optically active

BIOMOLECULES ⇒ (I) CARBOHYDRATES

Classification of Carbohydrates

- **Sugars** (→ Crystalline, → Sweet in taste, → Water soluble)
  - **Monosaccharides**
    - Triose (eg D. Glyceraldehyde)
    - Tetrose (eg D-Threose, D-Erythrose)
    - Pentose (eg Ribose, Arabinose)
    - Hexose
      - Aldohexose (eg Glucose, Mannose, Glactose etc)
      - Ketohexose (eg D. Fructose)
  - **Oligosaccharides**
    - Disaccharides eg Maltose, Sucrose, Lactose
    - Trisaccharide eg Raffinose
    - Tetrasacc
    - ⇒ upto Decasaccharide

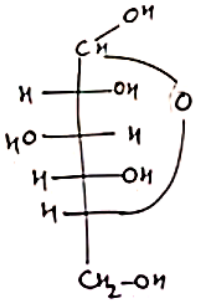
beyond 10, there are taken to be part of polysaccharides

- **Non Sugars** (→ Non crystalline, → Water insoluble) = not sweet
  - all polysaccharide
    - eg Starch, cellulose, glycogen etc

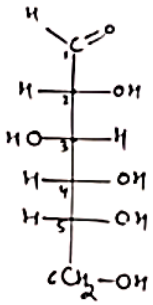
BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)

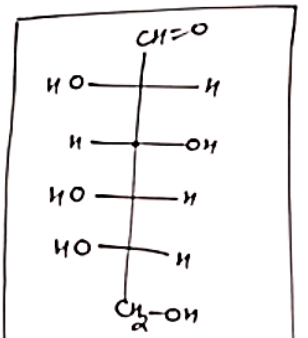
$[\alpha]_D = +52.7°$



Cyclic Structure of D. Glucose



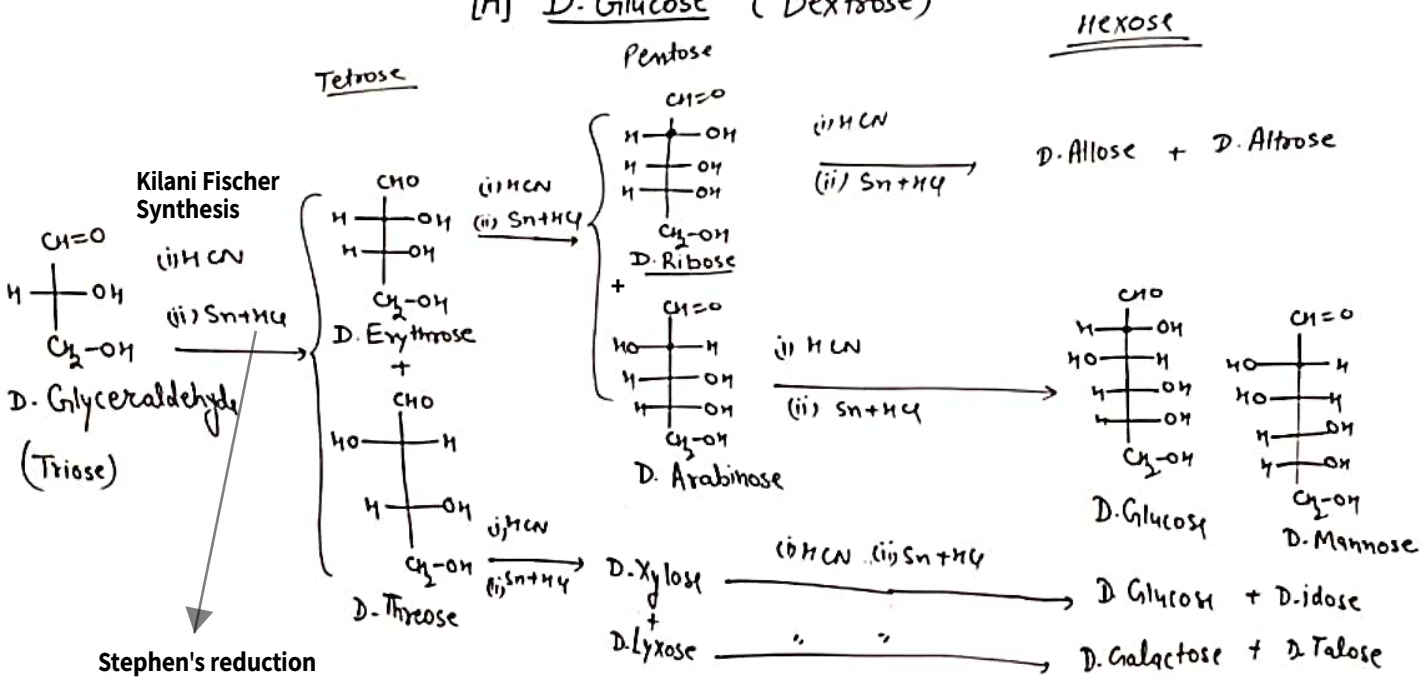
Open chain Structure of D. Glucose

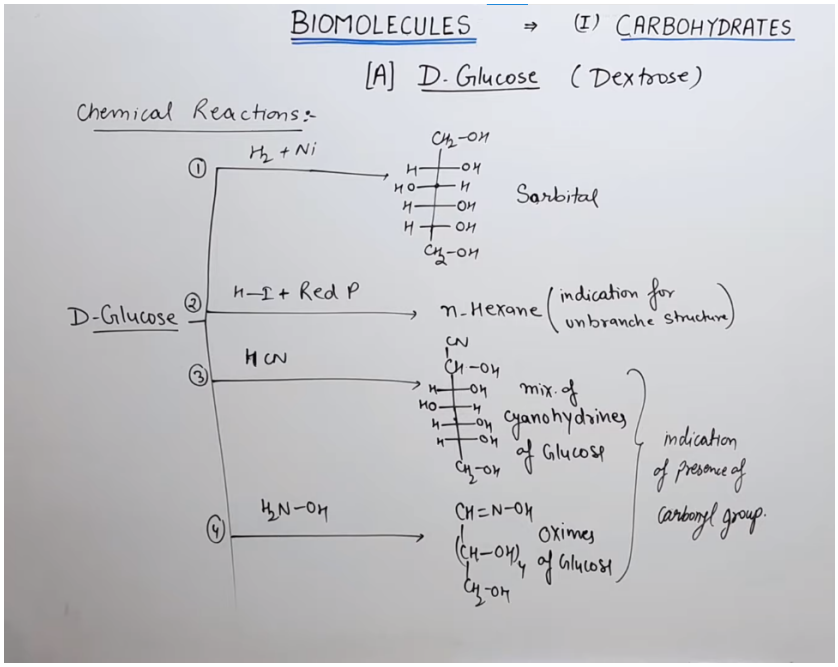


Open Chain Structure of L. Glucose

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)





BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)

Chemical Reactions:-

D-Glucose

⑤ ($Br_2 + H_2O$) mild oxidising agent
(R-CHO $\xrightarrow{Br_2+H_2O}$ R-COOH)
→ Gluconic acid

⑥ Tollen's Reagent → " + Ag↓

⑦ Fehling's solution → " + Cu_2O↓

⑧ Benect solution → " + Cu_2O↓

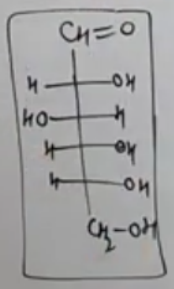
} indication of presence of -CHO group in glucose

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)

Chemical Reactions:-

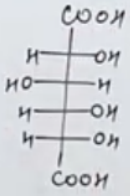
D-Glucose



(9) HNO_3 (oxidation)

$R\text{-}CHO \longrightarrow R\text{-}COOH$

$R\text{-}CH_2\text{-}OH \longrightarrow R\text{-}COOH$



Glucaric acid or Saccharic acid

(10) 2,4-DNP → No Reaction

(11) $NaHSO_3$ → No Reaction

(12) Schiff's Reagent → No Reaction

} Due to cyclic structure of Glucose

(13) Zymase enzyme (yeast) → $2C_2H_5OH + 2CO_2 + H_2O$

this rex. is used in liquor industries

maltose is first hydrolysed to glucose,
which is further reduced to ethyl alcohol by zymase
the formed ethyl alcohol is then recovered by distillation

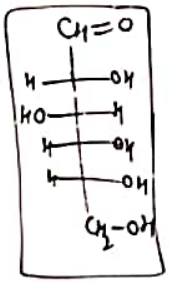
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BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)

Chemical Reactions:-

D-Glucose



(14) $5\ HIO_4 = H_5IO_6$ (Periodic acid) → $5\ H-COOH + CH_2=O$

(15) $5\ CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3 = Ac_2O$ → Penta acetyl-D-glucose (Glucose penta acetate)

($R-OH \xrightarrow{Ac_2O} R-OAc + R-O-\overset{O}{\overset{\|}{C}}-CH_3$)

(16) Me_2SO_4 → Pentyl methyl-D-glucose

($R-OH \xrightarrow{Me_2SO_4} R-O-CH_3$)

Indication of 5-OH groups

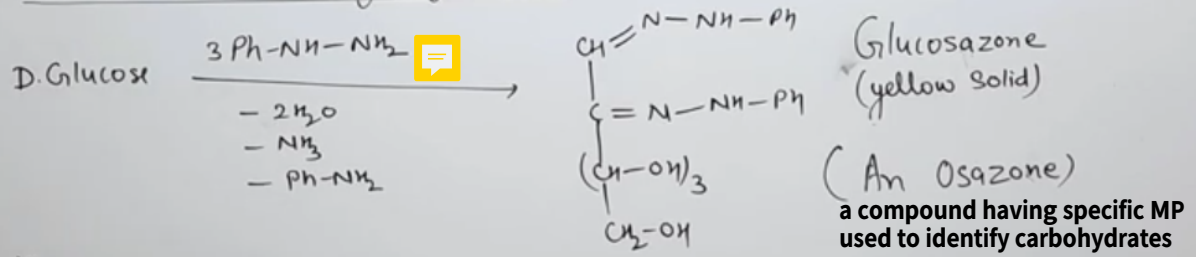
(A) MeOH + conc. HCl (or gas HCl) → Methyl-α-glucoside + Methyl-β-glucoside

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D. Glucose (Dextrose)

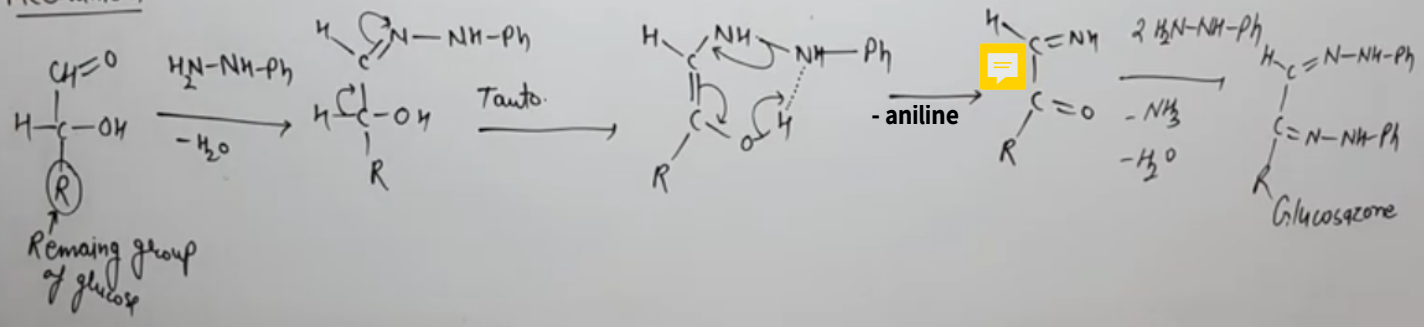
Chemical Reactions:-

(18) Reaction with phenyl hydrazine:-



a compound having specific MP
used to identify carbohydrates

Mechanism



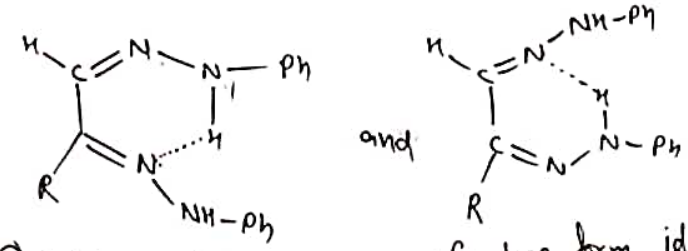
BIOMOLECULES ⇒ (I) CARBOHYDRATES

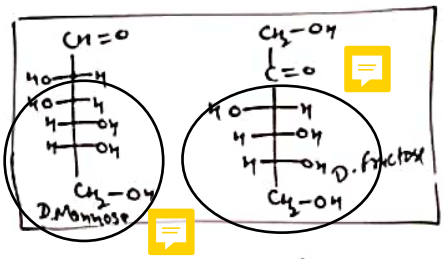
[A] D. Glucose (Dextrose)

Chemical Reactions:-

(1a) Reaction with phenyl hydrazine:-

Note ① Further reaction of osazone dose not possible with Ph-NH-NH_2 becaux osazone is stabilised by intramolecular H-Bonding





② D. Glucose, D-Mannose and D. fructose form identical osazone with Ph-NH-NH_2 due to identical R (Remaining group) (Same configuration)

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-

Cyclic Structure of glucose :-

Glucose exist in 6-membered cyclic hemiacetalic structure.

⇒ Evidence for cyclic structure of glucose.

1. Glucose does not react with 2,4-DNP, $NaHSO_3$ and Schiff's reagent.
2. Glucose forms two types of glucoside with methyl alcohol.
   ie. Methyl-α-glucoside and methyl-β-glucoside.
3. Glucose exist in two anomeric structures.
   ie (i) α-D-Glucose ⇒ M.P = 146°C, $[\alpha]_D = +112^\circ$
   (ii) β-D-Glucose ⇒ M.P = 148-150°C, $[\alpha]_D = +19^\circ$

⇒ **Anomer** :- Such type of diastereomer in carbohydres in which only one chiral carbon has diff. configuration (which is hemiacetalic / hemiketalic / acetalic / ketalic)

anomers exist only in cyclic structures

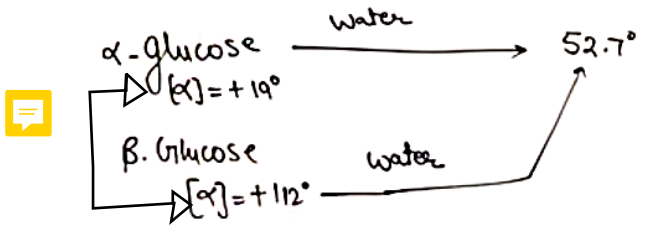
BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-

Cyclic Structure of glucose :-

4 Glucose shows mutarotation :-

→ Mutarotation :- The spontaneous change in specific rotation of an optical active compound upto equilibrium, called mutarotation.

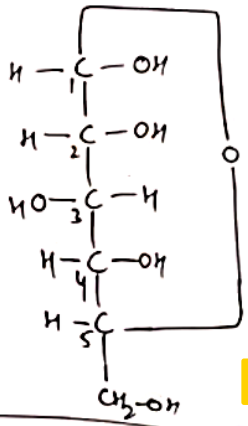


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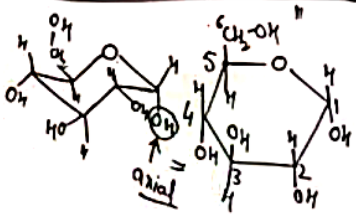
BIOMOLECULES ⇒ (I) CARBOHYDRATES

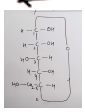
[A] D-Glucose :-

Cyclic Structure of glucose :-

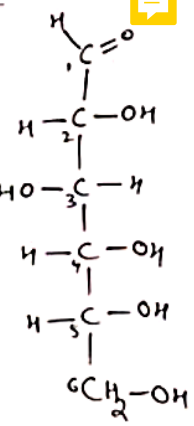


α-D-Glucopyranose

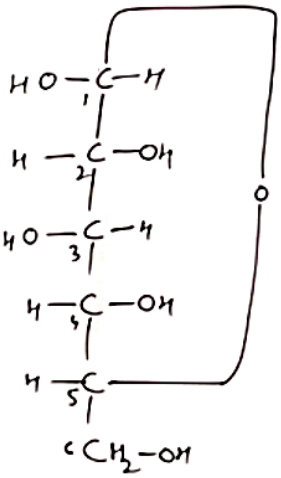




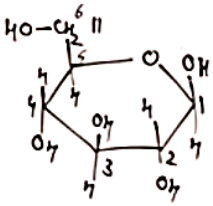
Haworth Structure ⇒

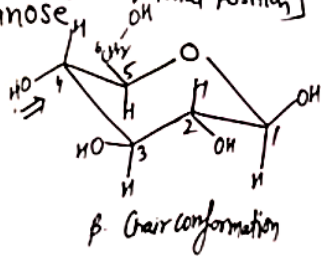


Open chain glucose



β-D-Glucopyranose





β Chair conformation

β-Glucose is more stable than α-glucose → all bulky groups on equitorial position

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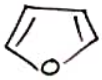
BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-

⇒ Glucose exist in Pyranose structure.



Pyrane



Furan

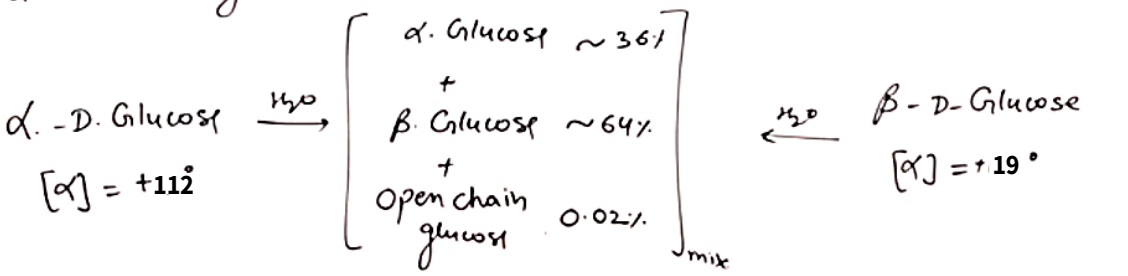
- **Pyranose :-**
  6-membered cyclic structure of carbohydrate
- **Furanose**
  5-membered, cyclic struc. of carbohydrate

⇒ Cyclic structure of glucose is not much stable. It can be opened by strong reagents like Tollen's reagent, Fehling sol., $H_2N\text{-}OH$, Br_2 water etc. but can not be opened by weak reagent like, 2,4 DNP, $NaHSO_3$ & Schiff's reagent.

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-

⇒ mutarotation in glucose

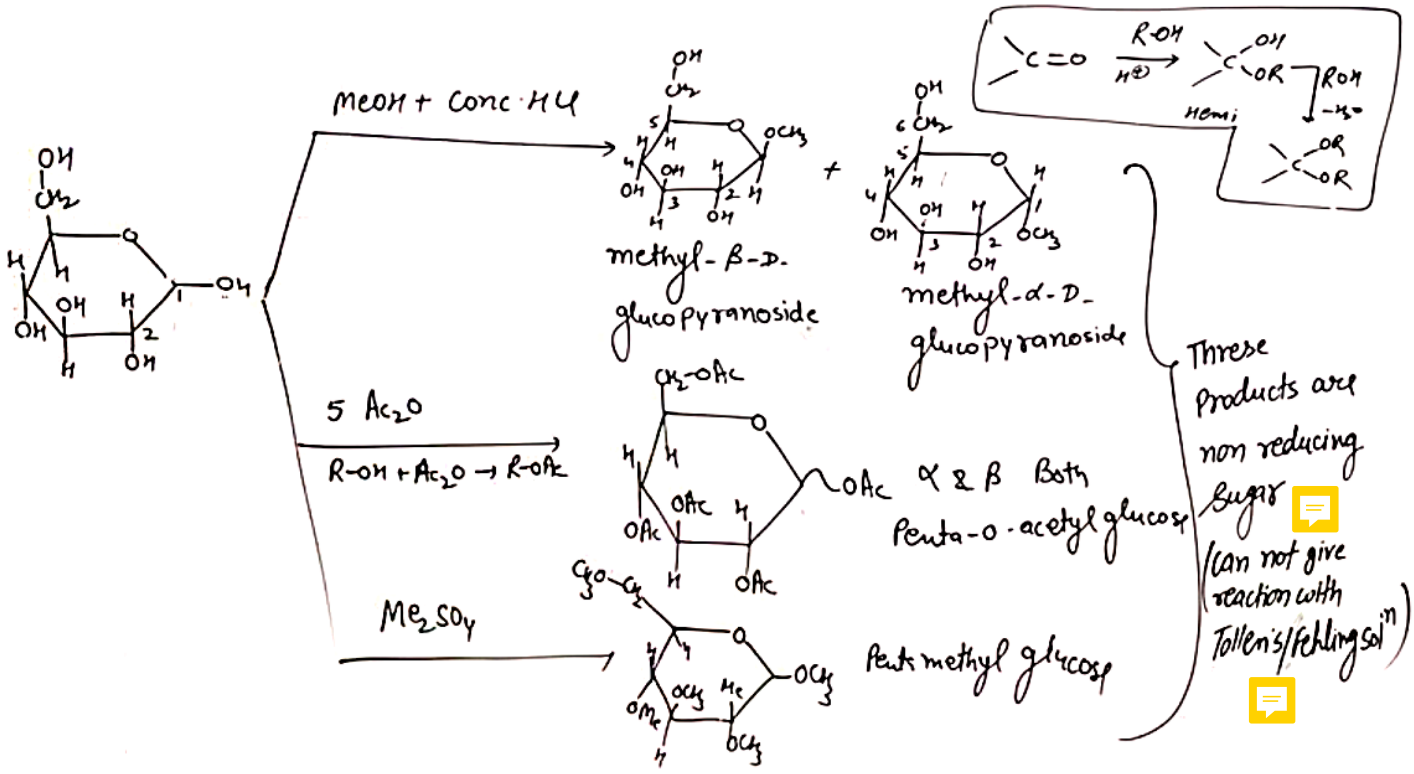


$$\frac{(64 \times 19) + (36 \times 112)}{100} \simeq 52^\circ$$

neglecting the optical rotation due to the open-chain form

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-



BIOMOLECULES ⇒ (I) CARBOHYDRATES

[A] D-Glucose :-

Difference between reducing and non reducing sugars

Properties	Reducing	Non reducing
① Free Hemiacetal or Hemiketal group	✓	X
② Reaction with		
- Tollen's reagent	✓	X
- fehling sol^{n}	✓	X
- Benedicts sol^{n}	✓	X
③ formation of Osazone	✓	X
④ Mutarotation	✓	X

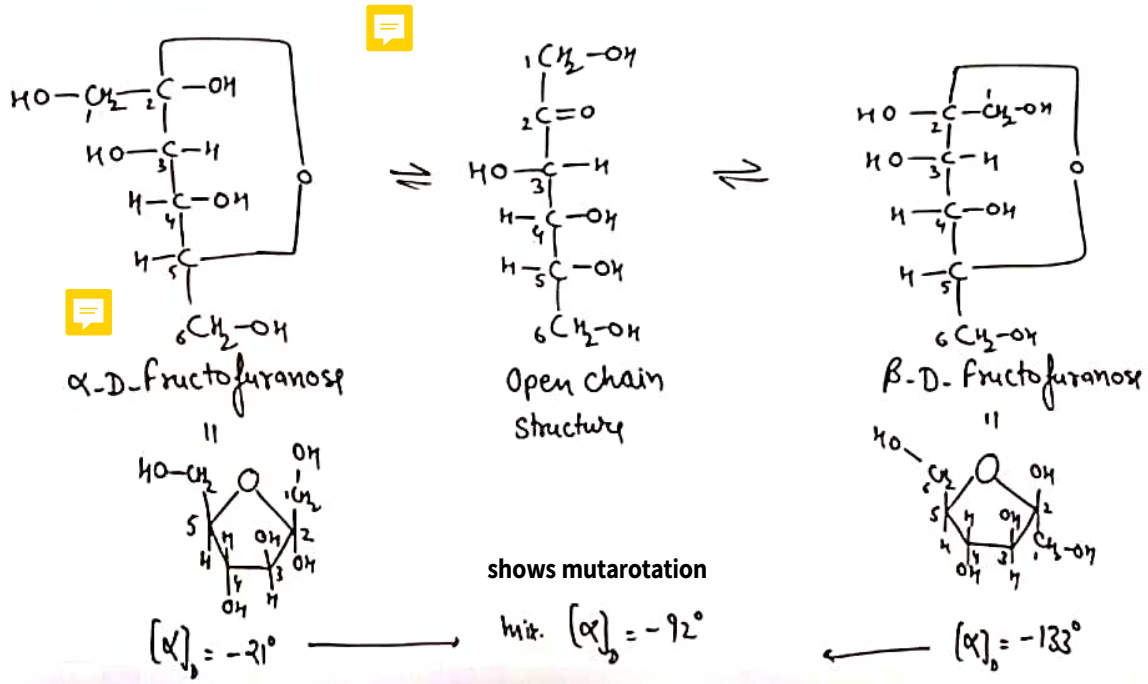
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BIOMOLECULES ⇒ (I) CARBOHYDRATES

[B] D. FRUCTOSE (Fruit sugar) = (Laevolose)

due to laevorotatory nature

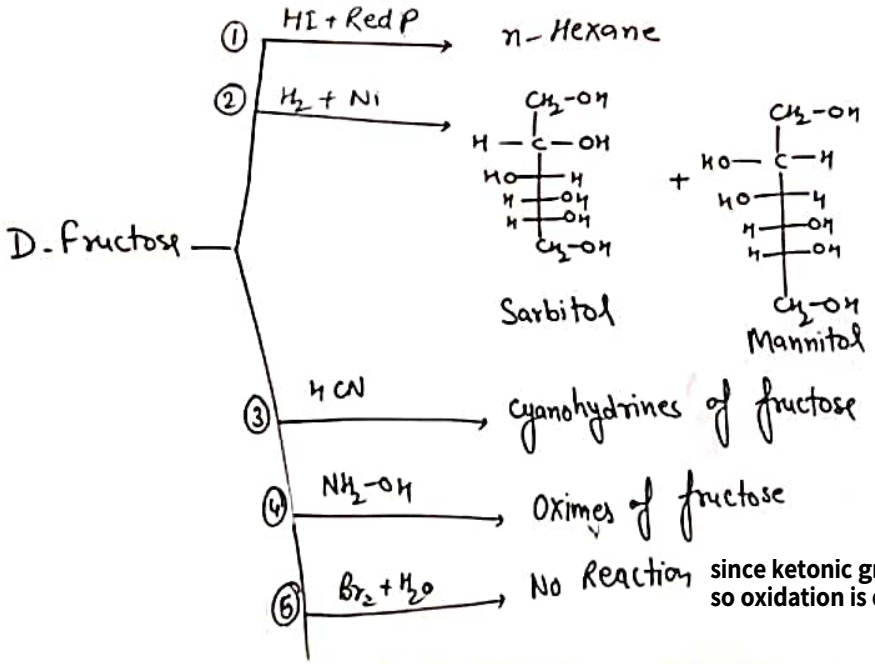
⇒ Ketohexose



BIOMOLECULES ⇒ (I) CARBOHYDRATES

[B] D. FRUCTOSE (Fruit sugar) = (Laevolose)

Chemical Reaction of fructose:-

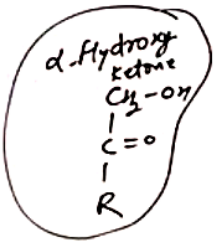


since ketonic grp.
so oxidation is difficult

BIOMOLECULES ⇒ (I) CARBOHYDRATES

[B] D. FRUCTOSE (Fruit sugar) = (Laevolose)

α-Hydroxy ketone



Chemical Reaction of fructose:-

D-Fructose (Reducing sugar)

⑥ Tollen's reagent → +ve test Ag↓

⑦ Fehling Sol^n or Benedict soln → +ve test + Cu_2O↓ Red

⑧ 2,4-DNP → No Rx^n

⑨ Schiff's Reagent → No Rx^n

⑩ $NaHSO_4$ → No Rx^n

⑪ $5Ac_2O$ → Penta-o-acetyl fructose

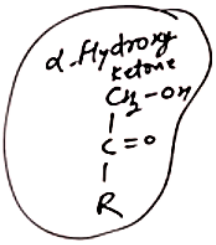
⑫ Me_2SO_4 → Penta-methyl-o-fructose

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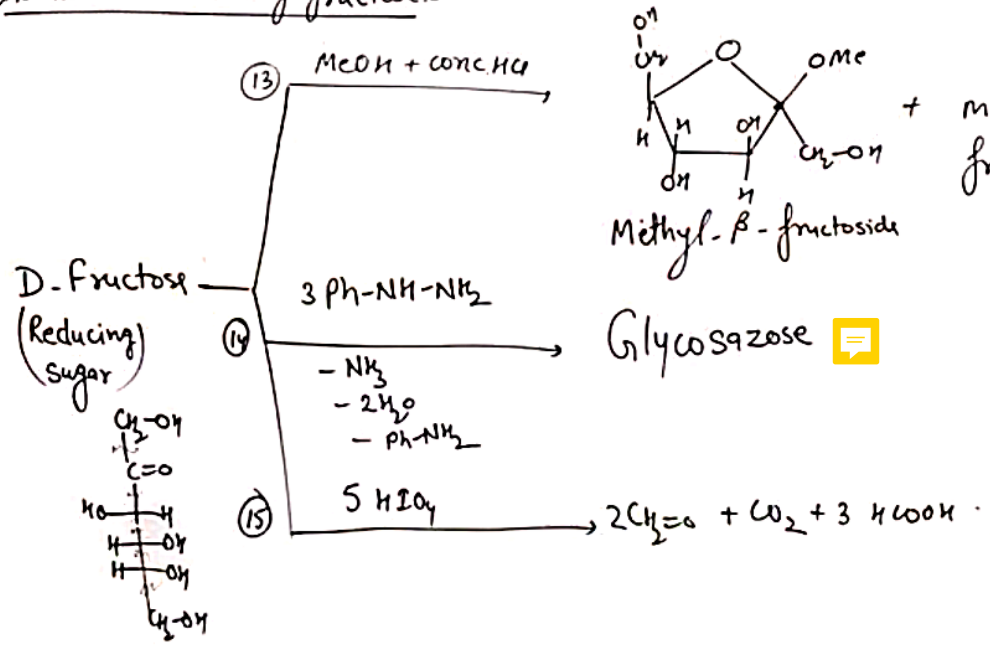
BIOMOLECULES ⇒ (I) CARBOHYDRATES

[B] D. FRUCTOSE (Fruit sugar) = (Laevolose)

α-Hydroxy ketone



Chemical Reaction of fructose:-



BIOMOLECULES (I) CARBOHYDRATES

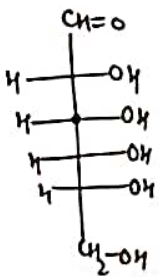
Epimer:- Such type of diastereomers in carbohydrate which have different configuration only on one chiral carbon (except anomeric carbon)

eg.1 D.Glucose and D.Mannose are C-2 epimer

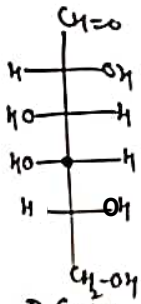
eg 2 D.Glucose and D-Allose .. C-3 .. ..

eg 3 D. Glucose and D. Galactose .. C-4 .. ..

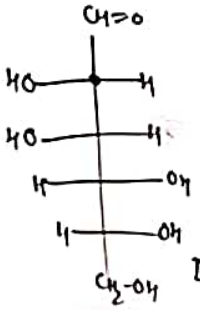
eg 4 D. Ribose and D. Arabinose .. C-2 epimer



D-Allose

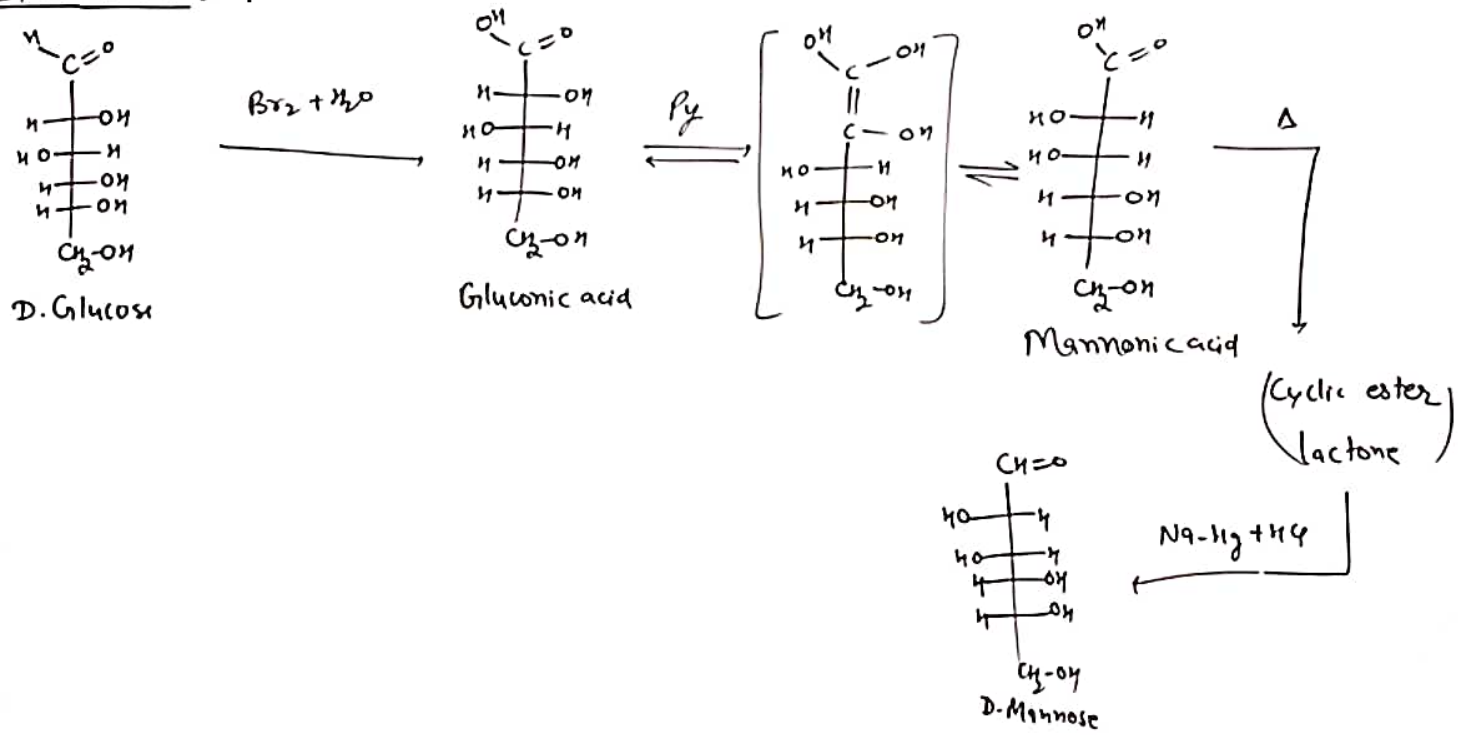


D. Galactose



D. Mannose

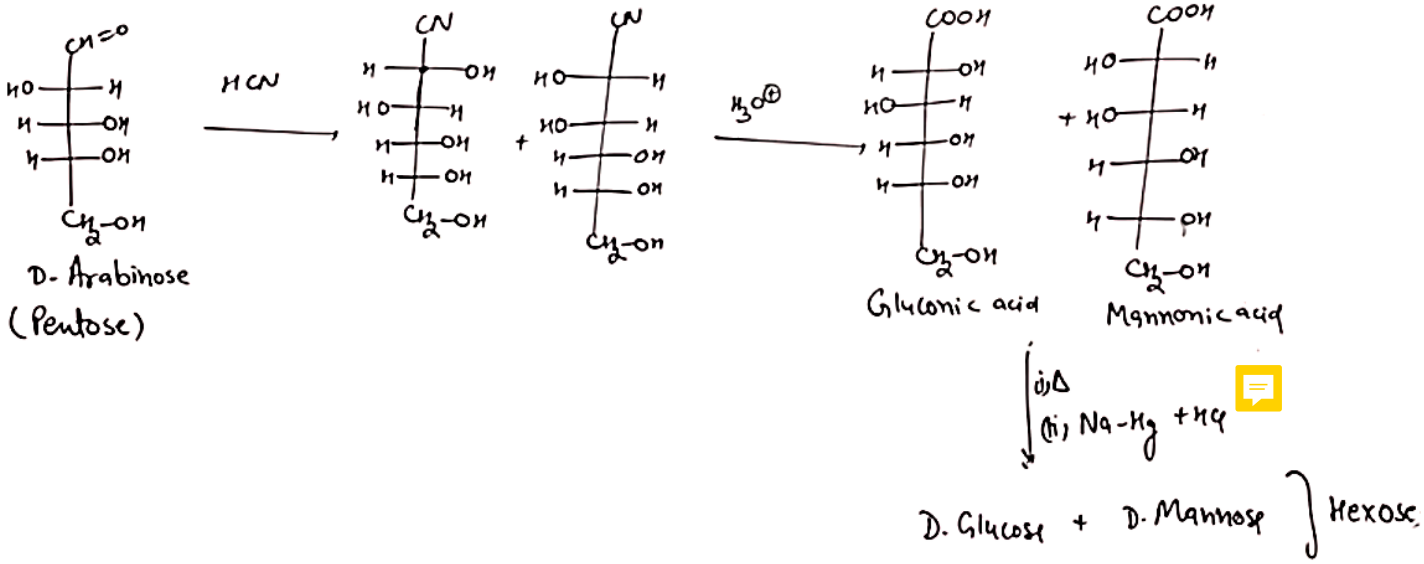
Epimerisation :- epimerisation of C2



BIOMOLECULES (I) CARBOHYDRATES

→ Lengthening of Aldose

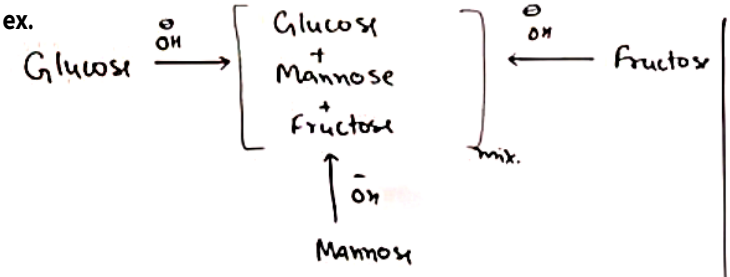
Killiani - Fischer synthesis

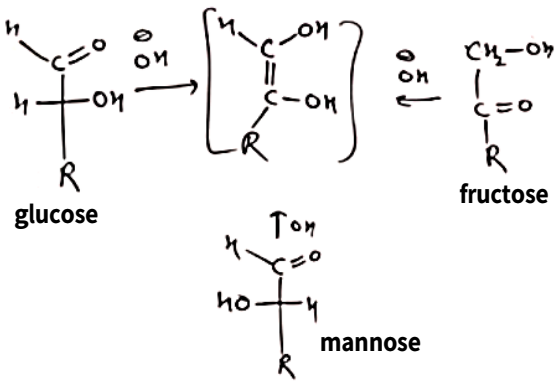


⇒ Lobry de - Bruyn - Van - Ekenstein transformation :-

In presence of dilute base, an isomerisation reaction takes place and form a mixture of Aldose, it's C-2 epimer and Ketose (like Glucose, Mannose and fructose) when any one is taken initially

ex.





R = remaining group

BIOMOLECULES (I) CARBOHYDRATES

Diasaccharide:- $(C_nH_{2n-1}O_{n-1})$—O—$(C_{n'}H_{2n'-1}O_{n'-1})$

combination of two mono-saccharides by ether linkage

- Non Reducing Diasaccharide
  - eg Sucrose (cane sugar)
- Reducing diasaccharides
  - eg Maltose (Malt sugar)
  - Lactose (Milk sugar)
  - Cellubiose

$C_{12}H_{22}O_{11}$ or $(C_6H_{11}O_5)$—O—$(C_6H_{11}O_5)$

BIOMOLECULES (I) CARBOHYDRATES

Diasaccharide:-

→ Hydrolysis of diasaccharide

(invert sugar)

① Sucrose + H_2O $\xrightarrow[\text{or}]{\text{dil HA (inversion of sugar)}}$ D. Glucose + D. fructose

$[\alpha] = +66.6°$ (invertase = sucrase) $[\alpha] = +52°$ $[\alpha] = -92°$

1 : 1

mix. $[\alpha]_{mix} = \frac{(+52°) + (-92°)}{2} = -20°$

② Maltose + H_2O $\xrightarrow[\text{or}]{H^{\oplus}}$ 2-D. Glucose

(Maltase)

③ Lactose + H_2O $\xrightarrow[\text{or}]{H^{\oplus}}$ D. Glucose + D. Galactose

(Lactase)

④ Cellubiose + H_2O $\xrightarrow{H^{\oplus}}$ 2-D. Glucose

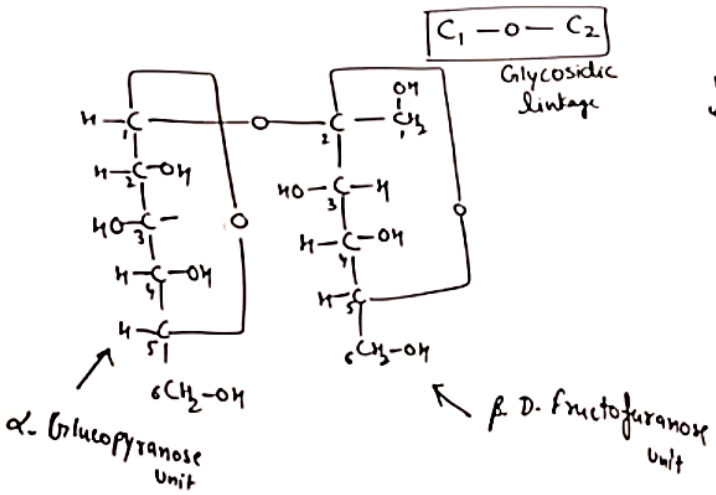
Subscribe

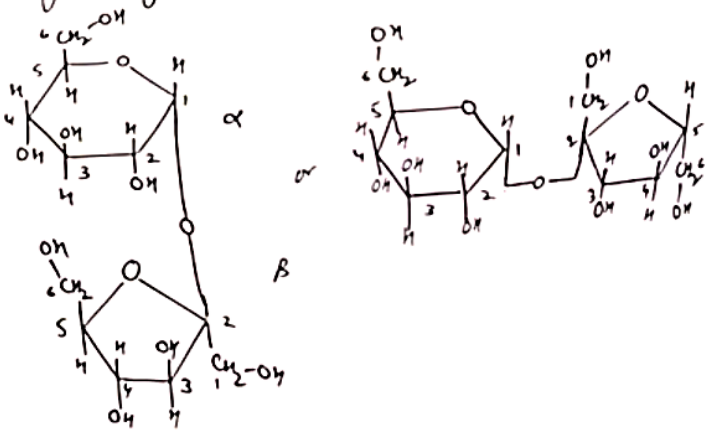
BIOMOLECULES (I) CARBOHYDRATES

Diasaccharide:-

① Structuse of sucrose:-

⇒ One α-D-Glucopyranose + One β-D-fructofuranose unit





isko upar laake ulta kr do

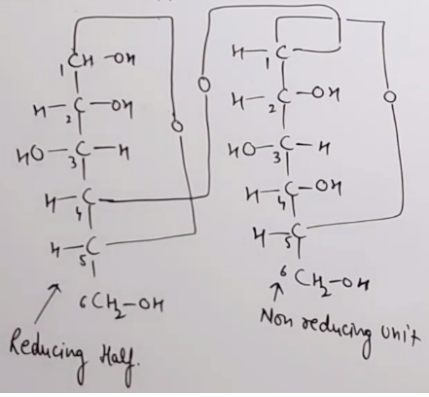
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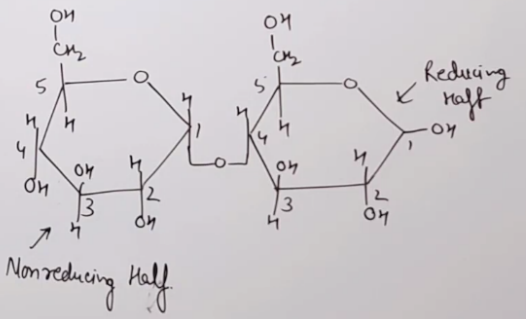
BIOMOLECULES (I) CARBOHYDRATES

Diasaccharide:-

② Structuse of Maltose

One Glucopyranose + one α-D-Glucopyranose ; α-C1-C4 glycosidic linkase.

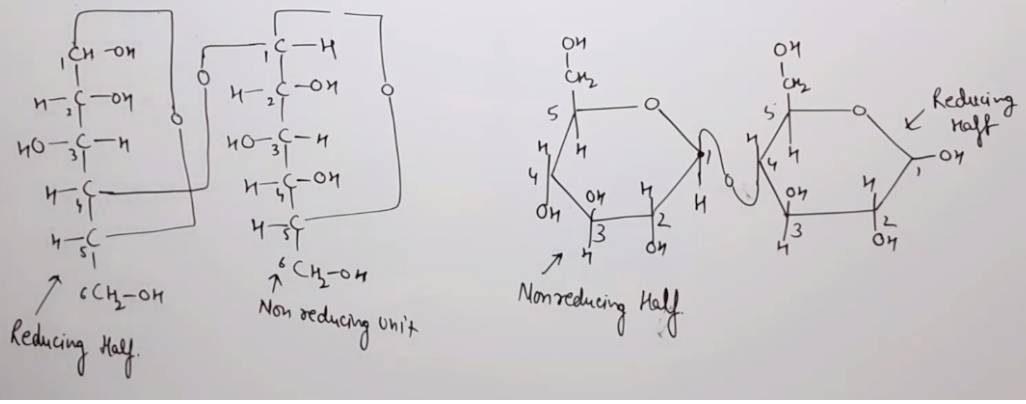




Diasaccharide:-

③ Structuse of Cellubiose

One Glucopyranose + one β-D-Glucopyranose ; β-C_1-C_4 glycosidic linkase.

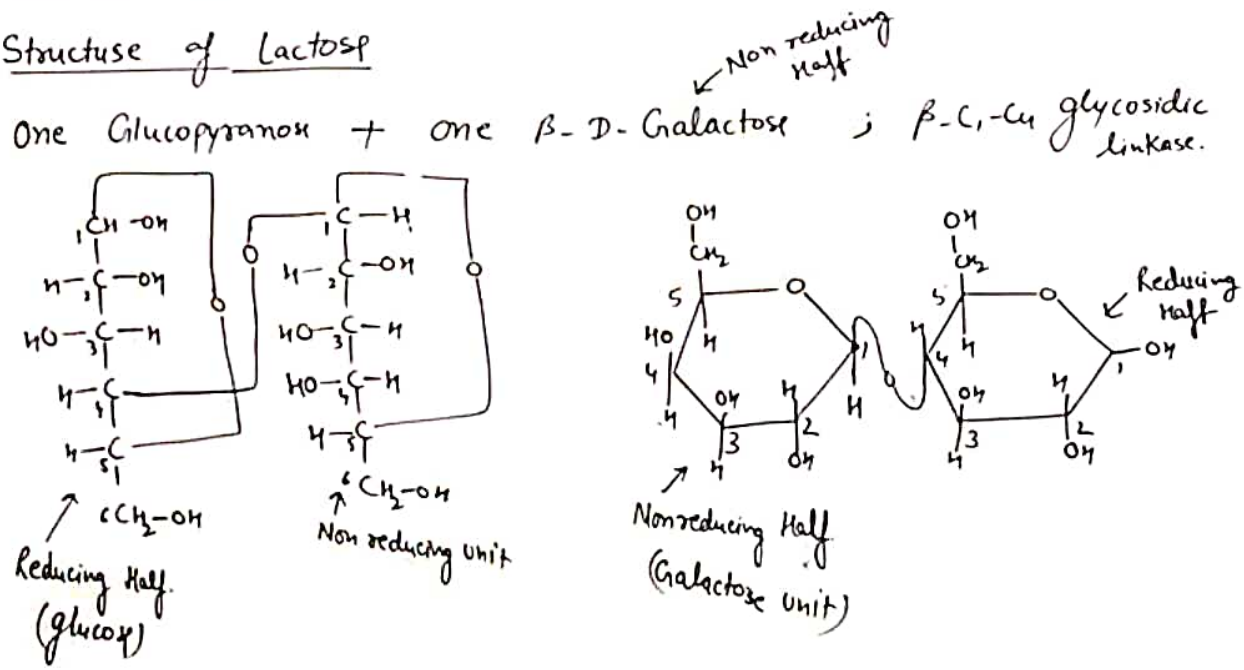


BIOMOLECULES (I) CARBOHYDRATES

Diasaccharide:-

④ Structuse of Lactose

One Glucopyranose + one β-D-Galactose ; β-C_1-C_4 glycosidic linkase.



BIOMOLECULES ⇒ (I) CARBOHYDRATES

POLYSACCHARIDES :-

⇒ Natural polymer of monosaccharides

⇒ All Polysaccharides are non reducing

① STARCH :-

⇒ Monomer unit of Starch ⇒ α-D-glucopyranose

⇒ Dimer " " " " ⇒ α-Maltose

⇒ with C_1–C_4 α-glycosidic linkage

Starch

- Amylose
- Amylopectine

BIOMOLECULES ⇒ (I) CARBOHYDRATES

POLYSACCHARIDES :-

① STARCH

Amylose

① 20% Part of starch
② Soluble in hot water
③ Lesser degree of polymerisation. (lesser glucose units)
④ lesser M.wt.
⑤ Blue complex form with I_2
⑥ Unbrached structure

Amylopectin

① 80% Part of starch
② insoluble in water
③ high degree of polymerisation. (higher glucose units.
④ high M.wt.
⑤ Violet complex with I_2
⑥ Brached chain structure.

⇒ (Branching starts from after every 25th to 30th glucose units.)
⇒ Braching present between C_1-C_6).

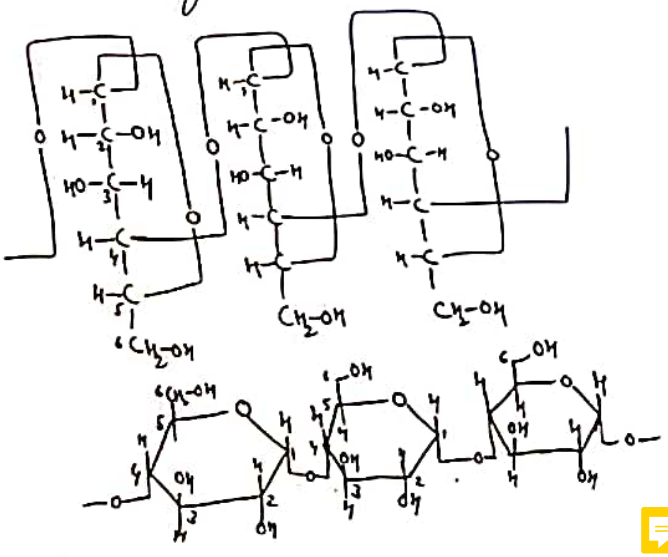
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BIOMOLECULES → (I) CARBOHYDRATES

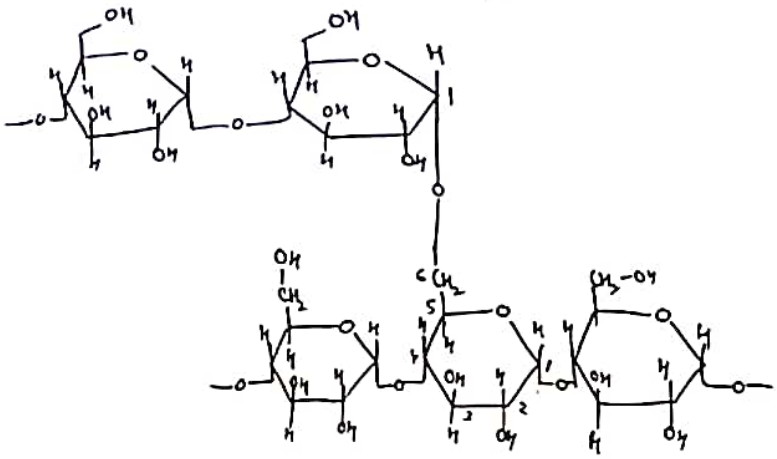
POLYSACCHARIDES :-

① STARCH

Amylose



Amylopectin

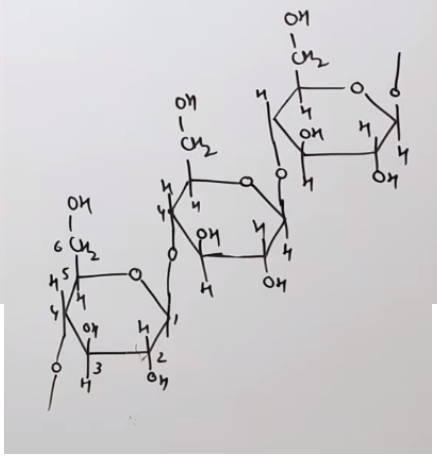


POLYSACCHARIDES :-

② Glycogen :- Structure of glycogen is same as of amylopectin but it has more branched structure than amylopectin and have branch at 5th - 6th glucose unit

③ Cellulose :-

- Natural polymer
- Monomer unit = β-D-Glucopyranose
- Dimer unit → β-Cellubiose
- Cellulose has unbranched structure
- higher degree of polymerisation than starch.



BIOMOLECULES → (II) AMINO ACIDS

- Proteins are natural polymer of amino acids (monomers)

Proteins $\xrightarrow{\text{Hydrolysis}}$ polypeptides ⟶ Peptides ⟶ Amino acids.

⇒ There are 20 natural amino acids exist in which 10 amino acids are essential

⇒ 'ine' ~~-ine~~ suffix

Essential amino acid

⇒ These are not synthesised in our body.

①	P	⇒ Phenyl alanine	⇒ Phe
②	V	⇒ Valine	⇒ Val
③	T	⇒ Threonine	⇒ Thr
④	T	⇒ Tryptophane	⇒ Trp
⑤	I	⇒ Isoleucine	⇒ I-Leu
⑥	M	⇒ Methronine	⇒ Met
⑦	H	⇒ Histidine	⇒ His
⑧	A	⇒ Arginine	⇒ Arg
⑨	L	⇒ Leucine	⇒ Leu
⑩	L	⇒ Lysine	⇒ Lys

Non essential amino acids

⇒ These are synthesised in our body

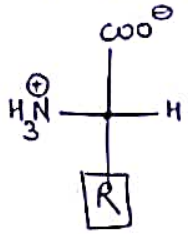
①	C	⇒ Cysteine	⇒ Cys
②	A	⇒ Alanine	⇒ Ala
③	A	⇒ Aspartic acid	⇒ Asp
④	A	⇒ Asparagin	⇒ Asn
⑤	G	⇒ Glycine	⇒ Gly
⑥	G	⇒ Glutamic acid	⇒ Glu
⑦	G	⇒ Glutamine	⇒ Gln
⑧	P	⇒ Proline	⇒ Pro
⑨	S	⇒ Serine	⇒ Ser
⑩	T	⇒ Tyrosine	⇒ Tyr

BIOMOLECULES ⇒ (II) AMINO ACIDS

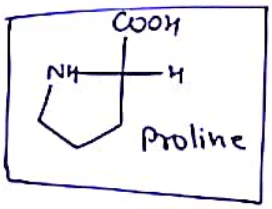
⇒ All natural amino acids are 'α- Amino carboxylic acid' except proline.

⇒ All natural amino acids are optically active with L- Raletive configuration except 'glycine'

⇒ General Structure



highly polar



Zwitter ionic (dipole structure)
Structure of amino acid

⇒ All natural amino acids are white crystalline solid.

⇒ They have high M.P., high density

⇒ Soluble in polar solvent but insoluble in non polar solvents.

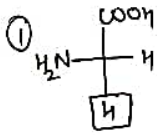
BIOMOLECULES ⇒ (II) AMINO ACIDS

STRUCTURES OF AMINO ACIDS:-

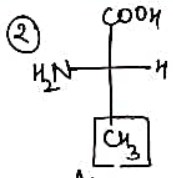
(I) Neutral amino acids:-

no. of $-NH_2$ = no. of $-COOH$

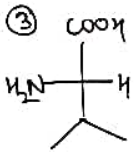
⇒ when R = Aliphatic (alkyl)



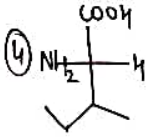
① Glycine



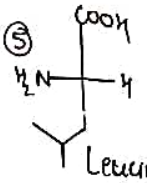
② Alanine



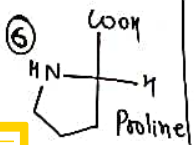
③ Valine



④ Isoleucine

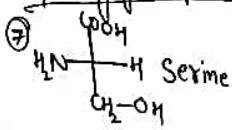


⑤ Leucine

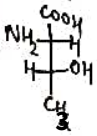


⑥ Proline

⇒ Hydroxy group containing amino acid

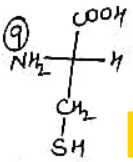


⑦ Serine

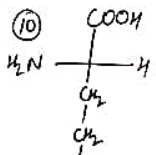


⑧ Threonine

⇒ S containing amino acids.

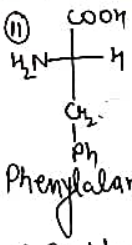


Cysteine

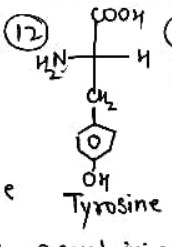


Metheonine

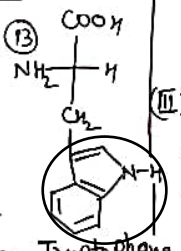
⇒ aromatic amino acid



Phenylalanine

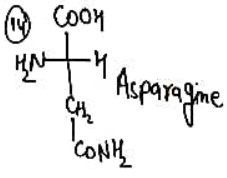


Tyrosine

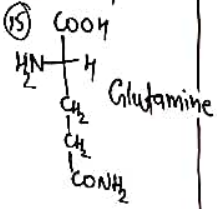


Tryptophane

⇒ amide group containing:-



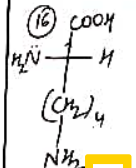
Asparagine



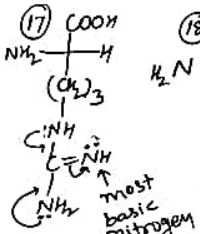
Glutamine

(II) Basic amino acid

no. of $-NH_2 > -COOH$

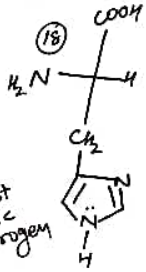


Lysine



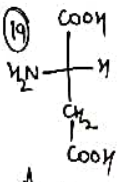
Arginine

(most basic amino acid)



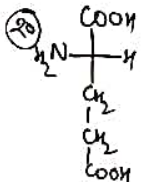
Histidine

(III) Acidic amino acid (−COOH > $-NH_2$)



Aspartic acid

(most acidic amino acid)

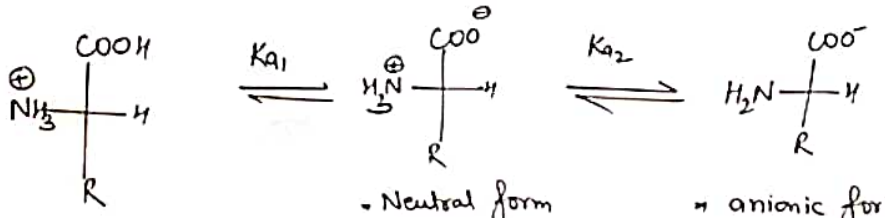


Glutamic acid

BIOMOLECULES ⇒ (II) AMINO ACIDS

DIPOLE STRUCTURE OF AMINO ACID :-

- For Neutral amino acid :-



- At low pH
  - ⇒ Cationic form
  - ⇒ moves towards Cathode in electric field.

- Neutral form
  - ⇒ zwitter ion.

$$pI = pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

- Isoelectric point
- Does not show migration towards any electrode

- ⇒ anionic form
- ⇒ at high pH
- ⇒ moves towards anode in electric field.

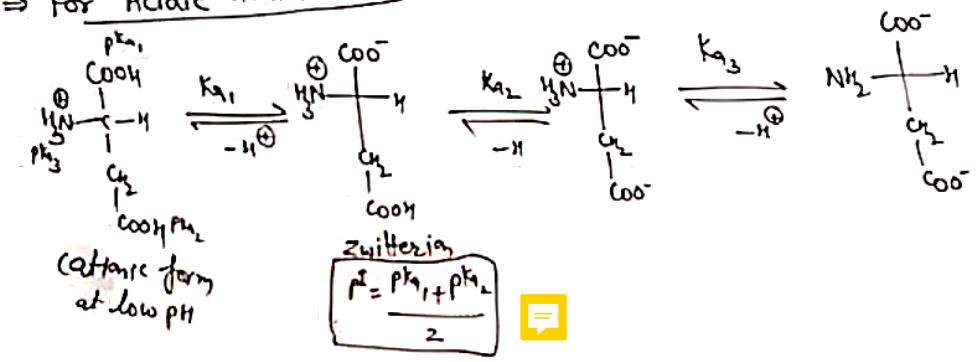
solubility of the amino acid in water is least in zwitter ionic form (i.e. at isoelectric point)

BIOMOLECULES ⇒ (II) AMINO ACIDS

DIPOLE STRUCTURE OF AMINO ACID :-

- $\Rightarrow$ P^I for neutral amino acid $= \dfrac{pK_{a_1} + pK_{a_2}}{2} \simeq$ 5.5 to 6.2
- $\Rightarrow$ P^I for acidic amino acid $= \dfrac{pK_{a_1} + pK_{a_2}}{2} \simeq$ 3 to 4
- $\Rightarrow$ P^I for basic amino acids $= \left[\dfrac{pK_{a_2} + pK_{a_3}}{2}\right] \simeq$ 10 to 11

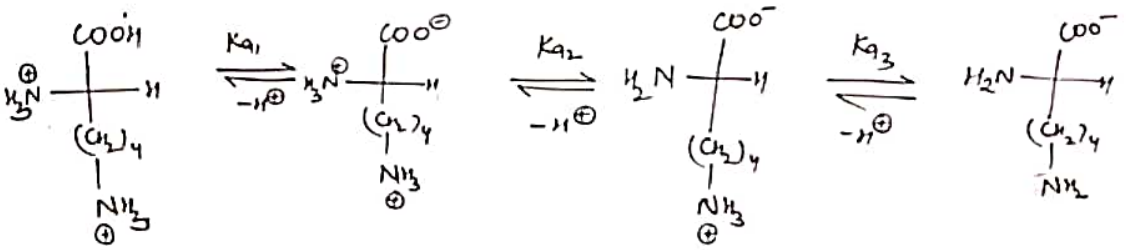
$\Rightarrow$ For Acidic amino acid



BIOMOLECULES ➧ (II) AMINO ACIDS

DIPOLE STRUCTURE OF AMINO ACID :-

→ For basic amino acids



Lysine
at low pH

$pK_1 = 2.18$
$pK_2 = 8.95$
$pK_3 = 10.79$

Zwitter

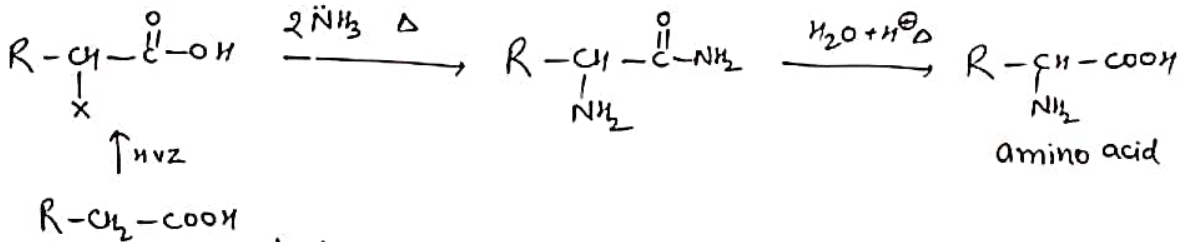
$$pI = \frac{pK_{a_2} + pK_{a_3}}{2}$$

$$pI = \frac{8.95 + 10.79}{2} = \frac{19.74}{2} = 9.9$$

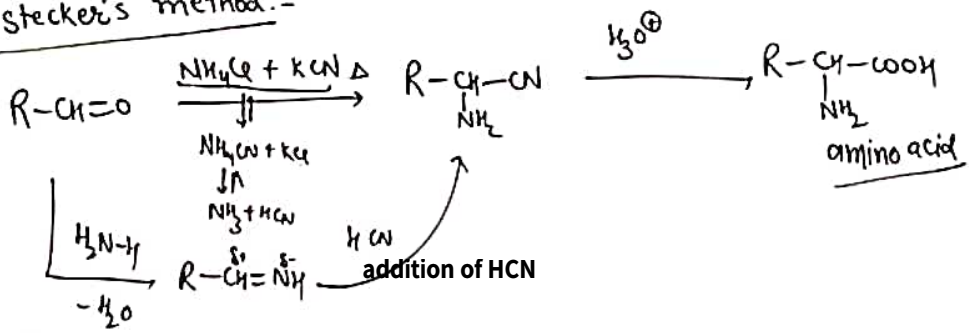
BIOMOLECULES ⇒ (II) AMINO ACIDS

METHODS OF FORMATION OF AMINO ACIDS:-

① By α-Halo acids:-



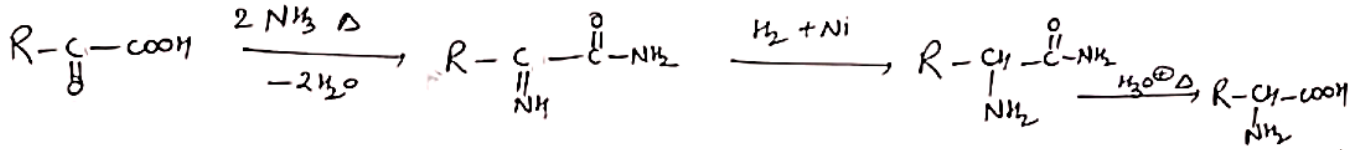
② By Strecker's method:-



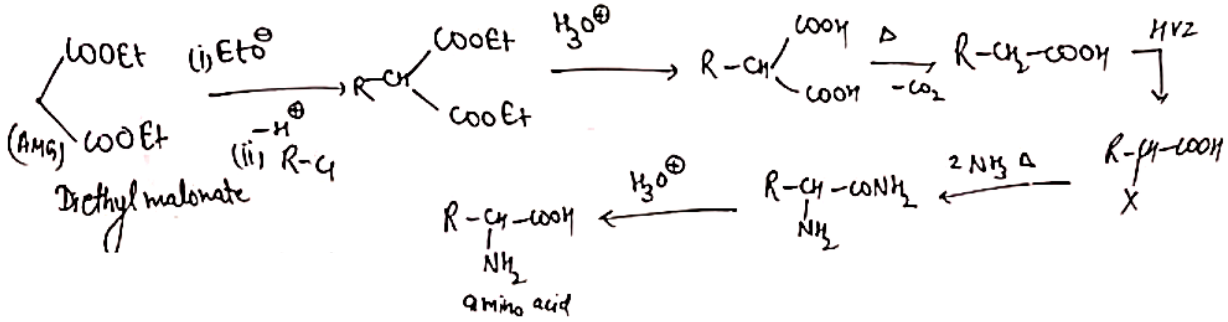
BIOMOLECULES ⇒ (II) AMINO ACIDS

METHODS OF FORMATION OF AMINO ACIDS:-

③ By α-Keto acid:- (KOOP Synthesis)



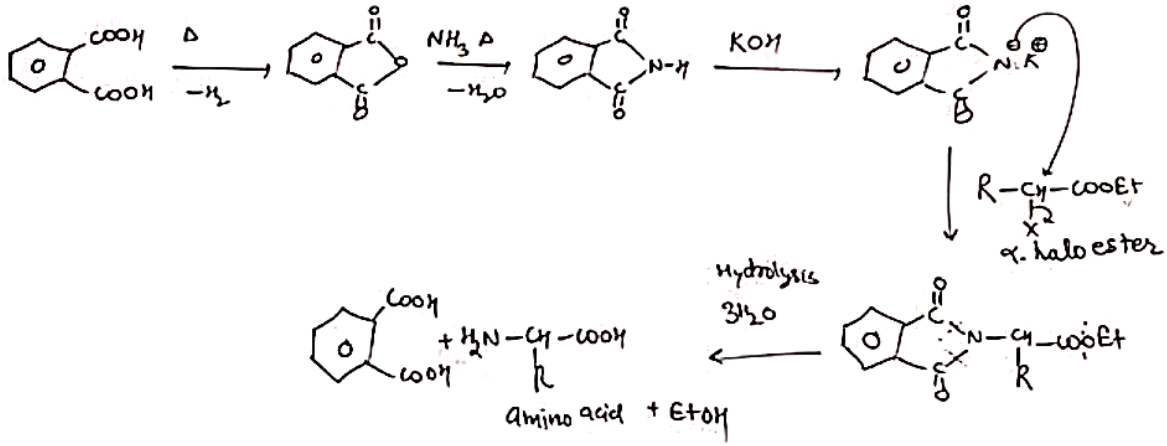
④ By Diethyl malonate:-



BIOMOLECULES ⇒ (II) AMINO ACIDS

METHODS OF FORMATION OF AMINO ACIDS:-

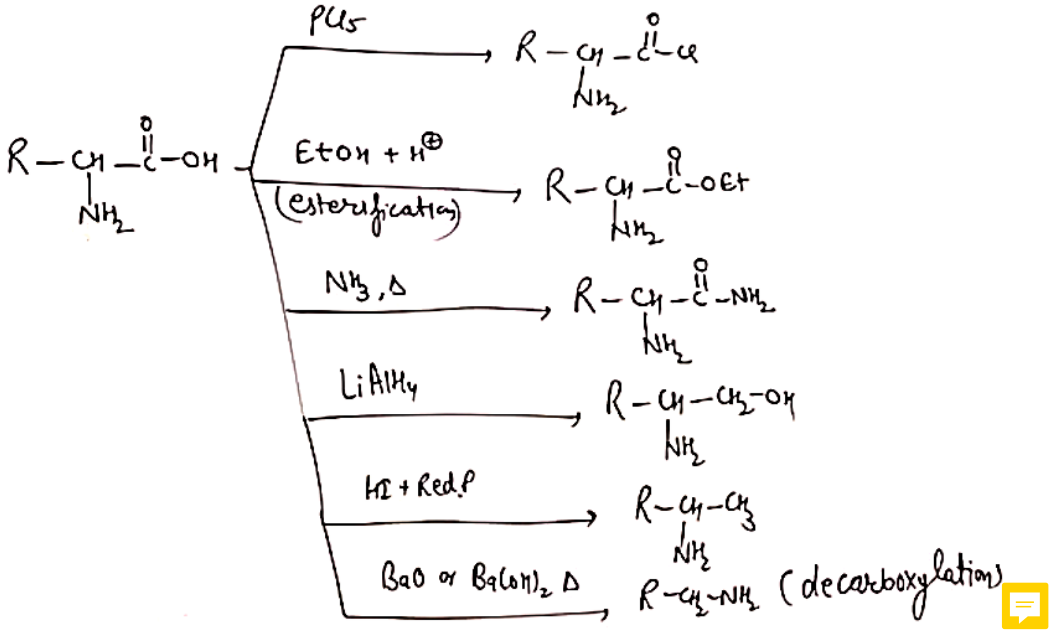
⑤ By Gabriel Phthalimide synthesis:-



BIOMOLECULES ⇒ (II) AMINO ACIDS

Chemical Reactions of amino acids:-

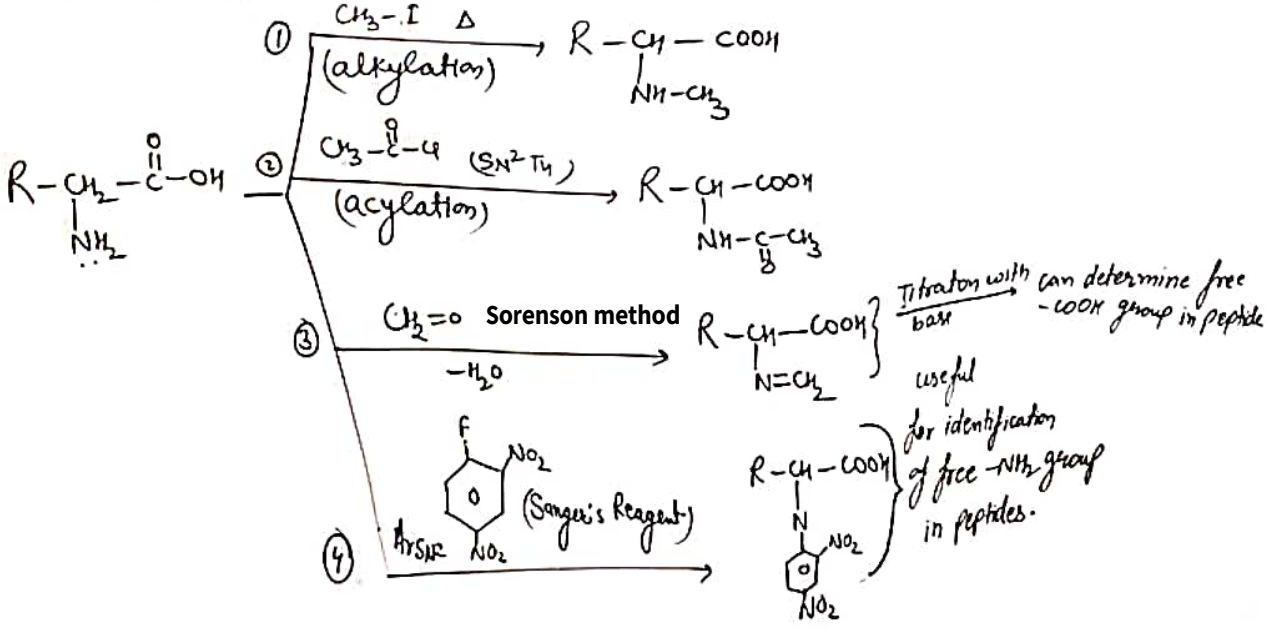
① Reactions of -COOH group:-



BIOMOLECULES ⇒ (II) AMINO ACIDS

Chemical Reactions of amino acids:-

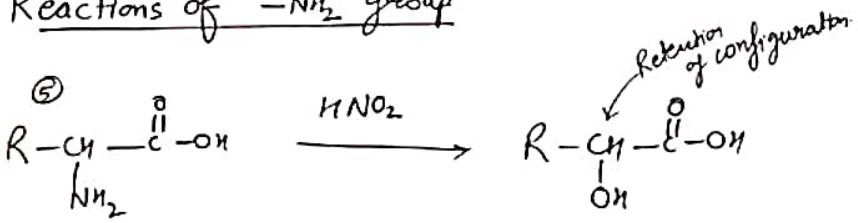
② Reactions of $-NH_2$ group

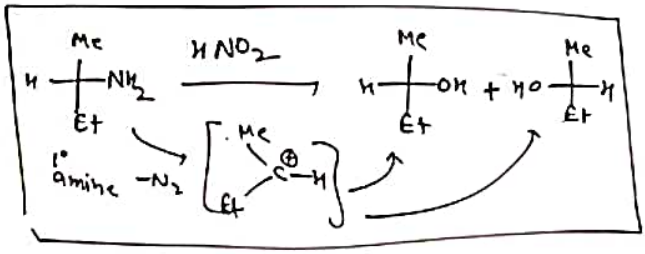


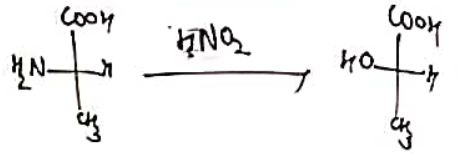
BIOMOLECULES ⇒ (II) AMINO ACIDS

Chemical Reactions of amino acids:-

② Reactions of $-NH_2$ group







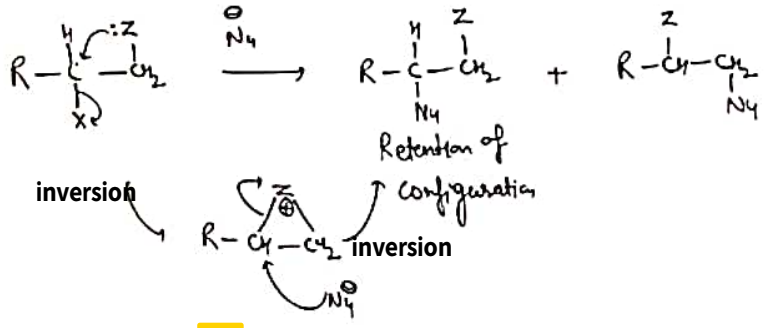
thus, always remember
rex. of amino acid with HNO2 only retention product is formed

BIOMOLECULES ⇒ (II) AMINO ACIDS

Chemical Reactions of amino acids:-

② Reactions of $-NH_2$ group

Neighbouring Group Participation (NGP)

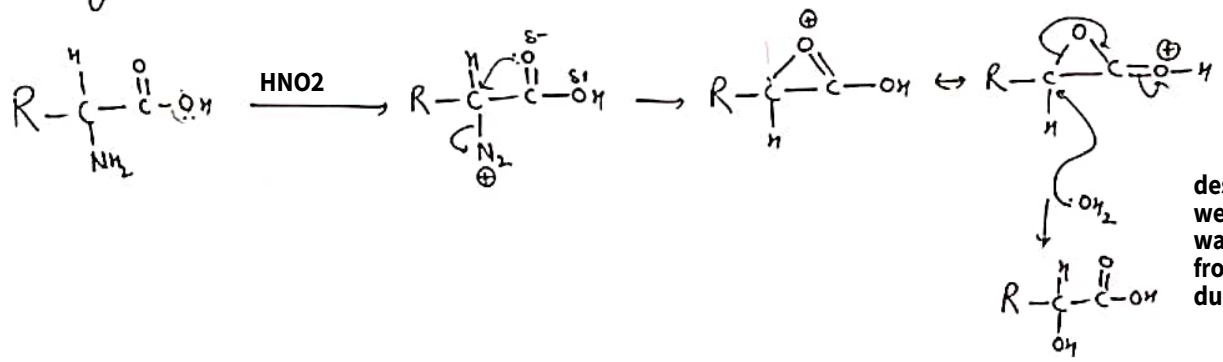


BIOMOLECULES ⇒ (II) AMINO ACIDS

Chemical Reactions of amino acids:-

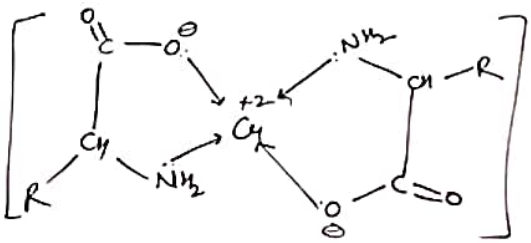
② Reactions of $-NH_2$ group

Neighbouring Group Participation (NGP)



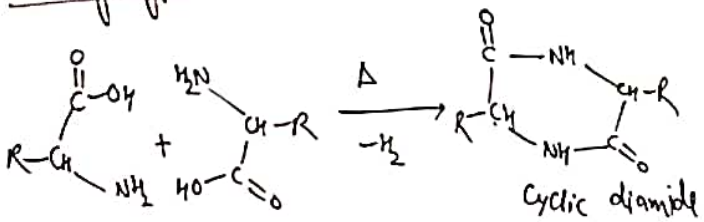
Chemical Reactions of amino acids:-

③ Reaction with Cu^{+2} salt:-

Amino acid + Cu^{+2} $\xrightarrow{\ominus OH}$ 

Blue

④ Heating effect:-



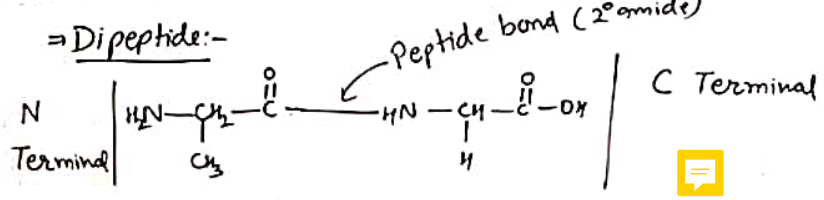
Diketopiparazine

Cyclic diamide

BIOMOLECULES ⇒ (II) AMINO ACIDS

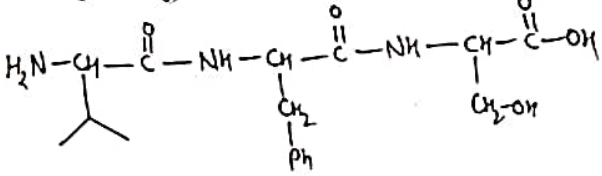
Peptides:- Condensation polymer of amino acids.

⇒ **Dipeptide:-**



Alanyl glycine

(Ala-Gly)

⇒  (Tripeptide)

Valyl-phenylalanyl-serine

(Val-Phe-Ser)

Q Ala + Gly → mix

Find no. of dipeptides

① Ala-Gly

② Gly-Ala

③ Gly-Gly

④ Ala-Ala

BIOMOLECULES ⇒ (II) AMINO ACIDS

PROTEINS :-

Proteins are considered as the biopolymers having a large number of amino acids linked to each other through peptide linkages containing 3D. Structure

⇒ Structures of Proteins :-
- (1°) Primary Structure of protein
- (2°) Secondary structure " "
- (3°) Tertiary " " "
- (4°) Quaternary " " "

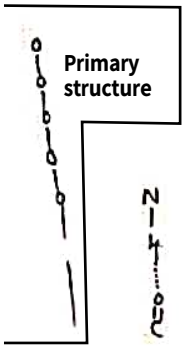
1. Primary Structure of protein :- The sequence of amino acids in a protein molecule is called primary structure of protein.

BIOMOLECULES ⇒ (II) AMINO ACIDS

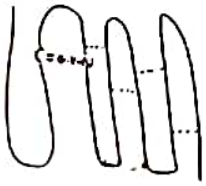
PROTEINS :-

(2) Secondary Structure of Protein :-

The conformation in which the polypeptide chains assume a shape as a result of H-bonding is called Secondary Structure of protein.



Types of 2° Structures

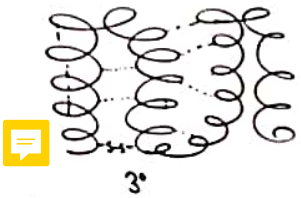
- α-Helix Structure
  - 
  - ~3.6 amino acid units in one round
  - ⇒ 13 MCS (13 member cyclic state (due to H-bonding))
  - eg α-keratin, myosin (α-keratin in hair, nails; myosin in muscles)
- 
- Sheet β-Pleated Structure
  - eg. Fibroin (Skin Protein)
  - Parallel β-pleated
  - Antiparallel β-pleated

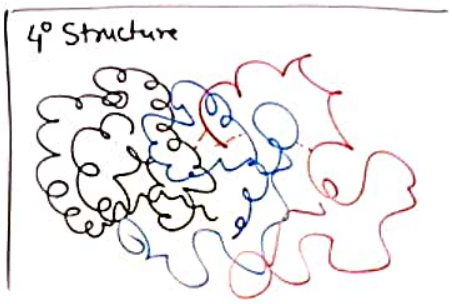
BIOMOLECULES ⇒ (II) AMINO ACIDS

PROTEINS :-

3. Tertiary Structure of Protein:-

The tertiary structure of proteins to its entire 3D structure i.e. the way in which the whole protein molecule fold up in the 3D space to form a particular shape.





4. Quaternary Structure of protein:-

Many proteins are composed of more than one polypeptide chain which are held together by some weak forces

The association of more than one polypeptide chain to form a stable large unit is called quaternary structure

BIOMOLECULES ⇒ (II) AMINO ACIDS

PROTEINS :-

⇒ Denaturation and renaturation of protein :-

The protein found in a biological system with a specific configuration and biological activity. is known as a native protein.

The lose of biological activity of a native protein due to physical and chemical treatment is called denaturation.

renaturation is the reverse of denaturation (i.e regeneration of the biological activities of the native protein). in most cases, denaturation is irreversible; only for few proteins renaturation is possible

BIOMOLECULES ⇒ (II) AMINO ACIDS

PROTEINS :-

Classification of proteins

(A) On the basis of molecular structure

① Fibrous proteins

When polypeptide chain run parallel and form fiber like (linear) structure called Fibrous Protein.

⇒ These protein chain held together by hydrogen bond and disulphide linkage.

⇒ Water insoluble

⇒ eg. Keratin, myosin

② Globular protein

When polypeptide chain is folded to form spheroidal shape it is called globular protein.

⇒ Water Soluble

⇒ eg. Albumin in egg. Enzyme etc.

BIOMOLECULES ⇒ (II) AMINO ACIDS

PROTEINS :-

Classification of proteins

(B) On the basis of chemical composition

① Simple protein

They give only amino acids on hydrolysis.

eg Albumin, Globulin, Glutalin in wheat, rice etc.

② Conjugate protein

In which protein part is combined with non protein part. These proteins give a non protein part with amino acids on hydrolysis

⇒ This non protein part is called prosthetic group

⇒ Prosthetic group can control the biological functions of Protein.

⇒ Prosthetic group may be carbohydrate, Phosphate, lipid etc.

③ Derived protein

When proteins hydrolysised by acid/ base/ enzymes then the degraded proteins called derived protein.

BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

- Nucleic acids are long chain polymer of nucleotides (Polynucleotide).
- Nucleic acids contain a polyphosphate ester chain.
- There are mainly two types of nucleic acids ① DNA (Deoxyribonucleic acid)
  ② RNA (Ribonucleic acid)

⇒ DNA is localised largely in the nucleus inside the chromose and small amount of DNA is also available in the cytoplasm (in mitochondia and chloroplasts)

H_3PO_4

⇒ RNA is also found both in nucleus and in cytoplasm.

⇒ DNA as the chief source of genetic information is copied in RNA molecule (Trascription)

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BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

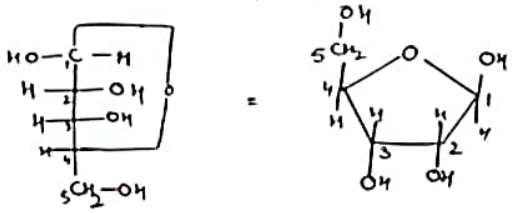
Chemical composition of Nucleic acids :-

→ Nucleic acid —Complete hydrolysis→ (Pento Sugar) + [Nitrogous Basis] + Phosphoric acid (H_3PO_4)

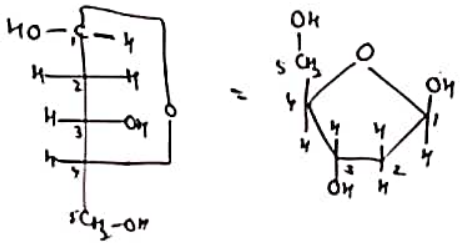
(Base – Sugar – Phosphate)

⇒ (1) Pentose sugar

- In RNA ⇒ β. D-Ribose



- In DNA = deoxy-β-ribose



BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

Chemical composition of Nucleic acids :-

→ Nucleic acid —Complete hydrolysis→ (Pento Sugar) + [Nitrogous Basis] + Phosphoric acid (H_3PO_4)

Base – Sugar – Phosphate

② Nitrogenous bases

- Purines (bicyclic rings)
  - Adenine (A)
  - Guanine (G)
- Pyrimidens (monocyclic rings)
  - Uracil
  - Cytosine
  - Thyamine

BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

Chemical composition of Nucleic acids :-

→ Nucleic acid —Complete hydrolysis→ (Pento Sugar) + [Nitrogous Bases] + Phosphoric acid (H_3PO_4)

(Base – Sugar – Phosphate)

② Nitrogenous bases

in DNA	in RNA
A ..... T	A ..... U
G C	G C

always pyrimidines form H bonds with purines

they cannot form H-bonds with each other (purine with purine or pyrimidine with pyrimidine is not possible)

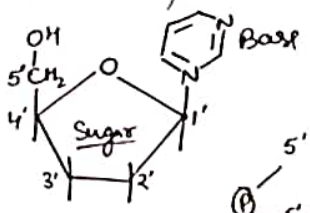
A always forms 2 H-bonds with T/U
G forms 3 H bonds with C

BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

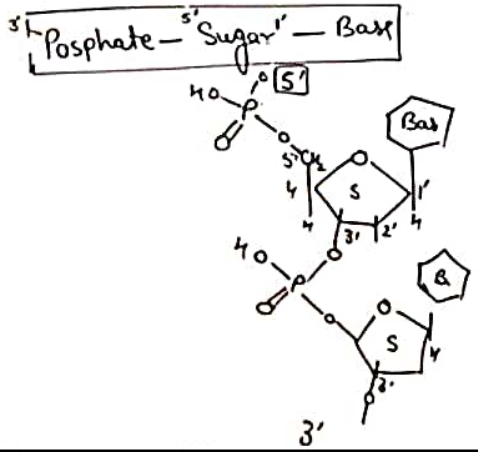
Chemical composition of Nucleic acids :-

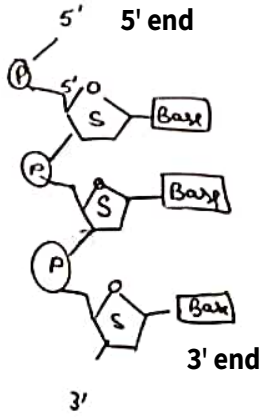
Nucleoside :- Sugar — Nitrogenous Base

β-N. Glycosidic linkage



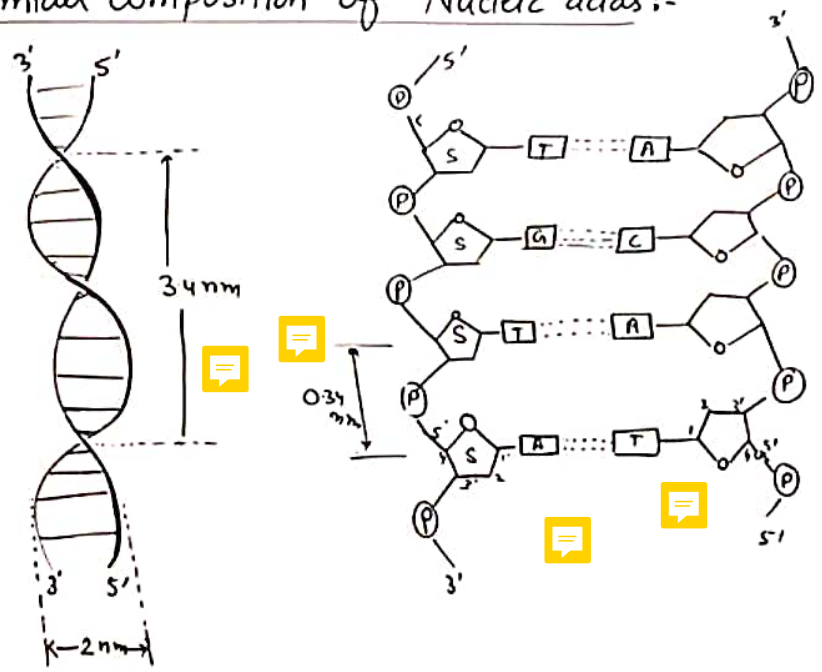
Nucleotide :- Posphate — Sugar — Base





BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

Chemical composition of Nucleic acids :-



Cargaff's Rule

Pur Pyr

no. of A = T

G = C

$\frac{A}{G} = \frac{T}{C}$ = specific value for diff. species

$\frac{A+T}{G+C}$

BIOMOLECULES ⇒ (III) NUCLEIC ACIDS

Types of RNA :-

① Ribosomal RNA (r-RNA) ⇒ 80%

② Messenger RNA (m-RNA) ⇒ 15%

③ Transfer RNA (t-RNA) ⇒ 5%

